Baterías y servicios complementarios:

El caso de Gran Bretaña

Luis Badesa

Dpto. de Ingeniería Eléctrica, UPM

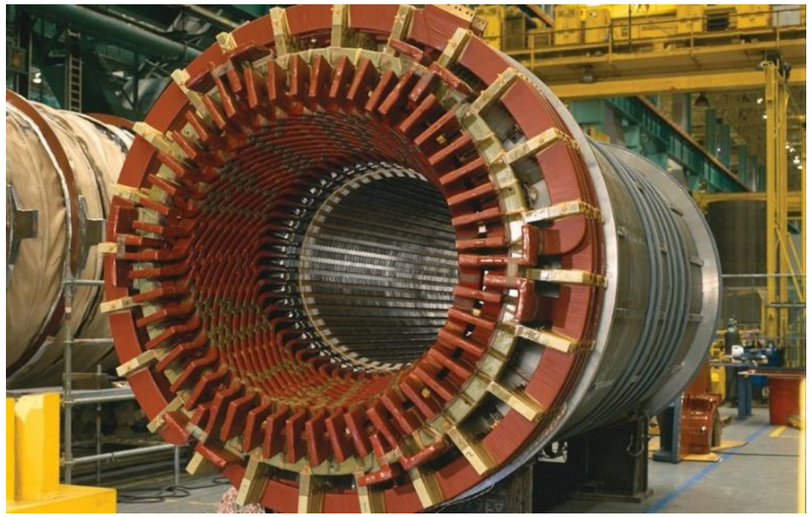
UNIVERSIDAD POLITÉCNICA DE MADRID

25 de septiembre 2025

escuela técnica superior de ingeniería y diseño industrial

Context: **Lower inertia** on the road to **lower emissions**

Thermal generators
(nuclear, gas, coal…)



Decarbonization

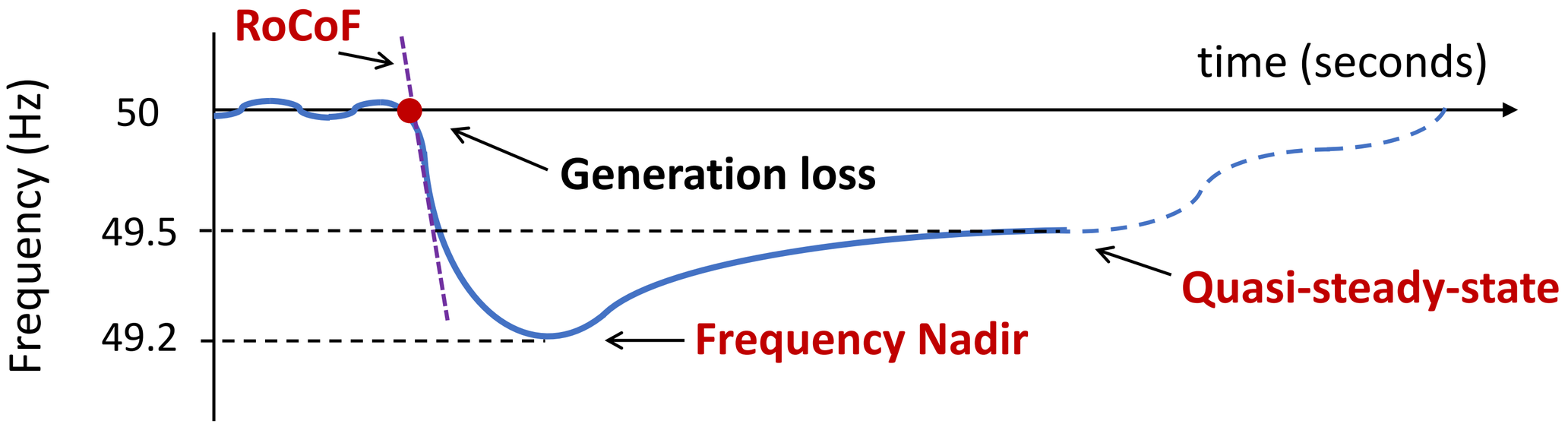
Most **renewables**:
no inertia



The **risk of instability has increased!**

Inertia <u>stores kinetic energy</u>:
this energy gave us time to contain a sudden generation-demand imbalance

Frequency stability

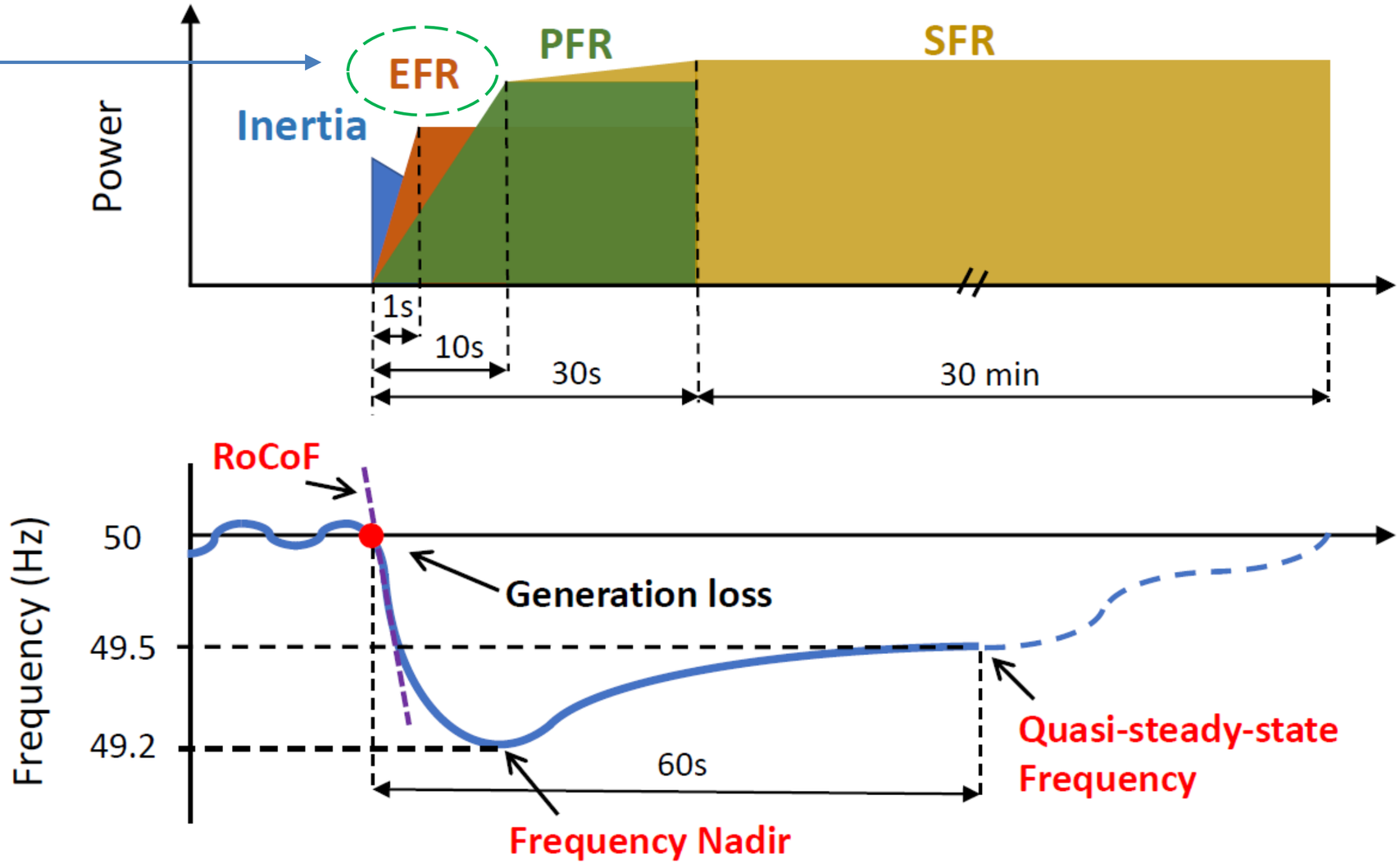


Key to keep frequency within safe limits to
avoid demand disconnection!

Frequency-containment services in GB

Enhanced Frequency Response, created in **2016** (currently named 'Dynamic Containment')

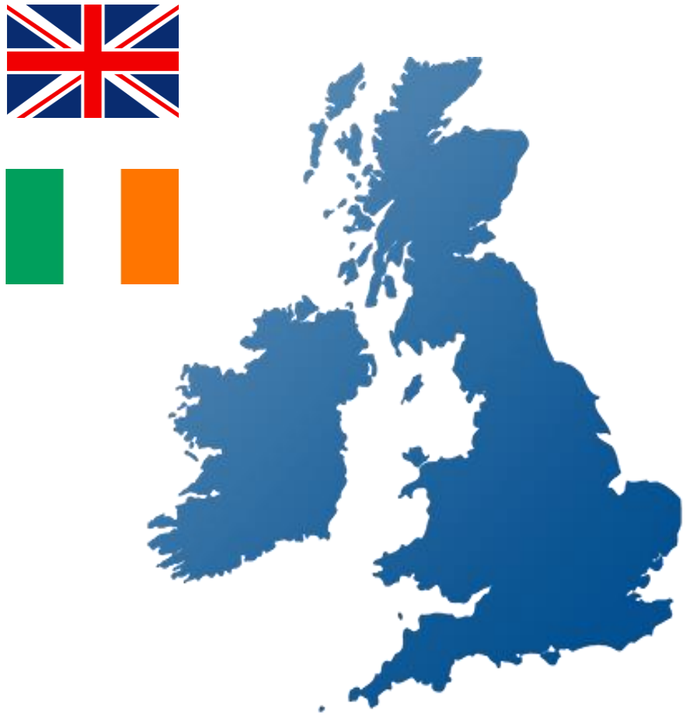
Solely provided by **battery** storage



Countries where this matters most

Islands, either geographical or electric islands like Texas

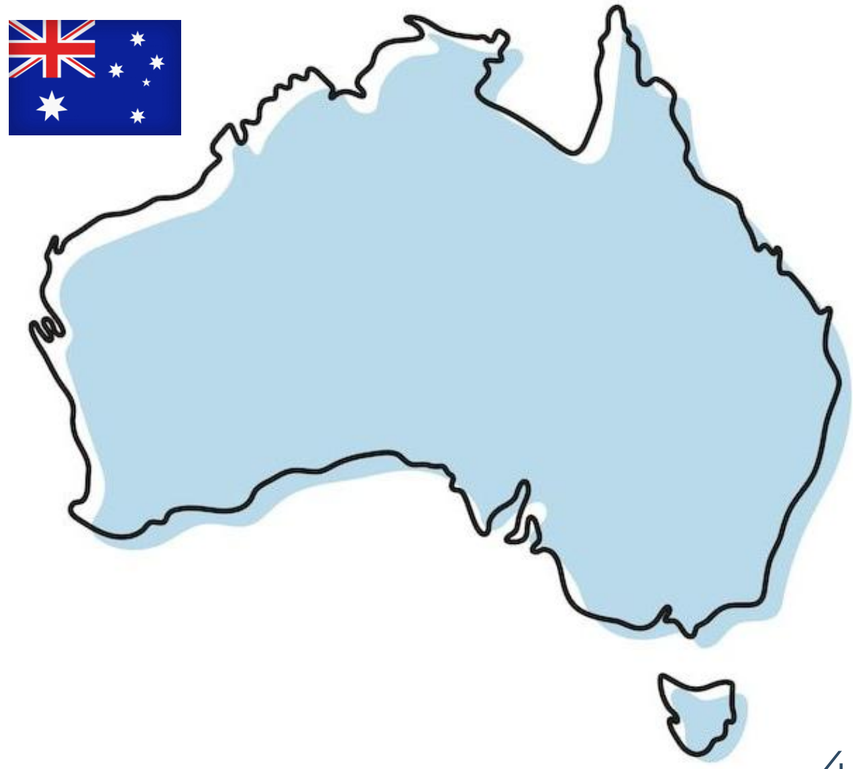
UK, Ireland



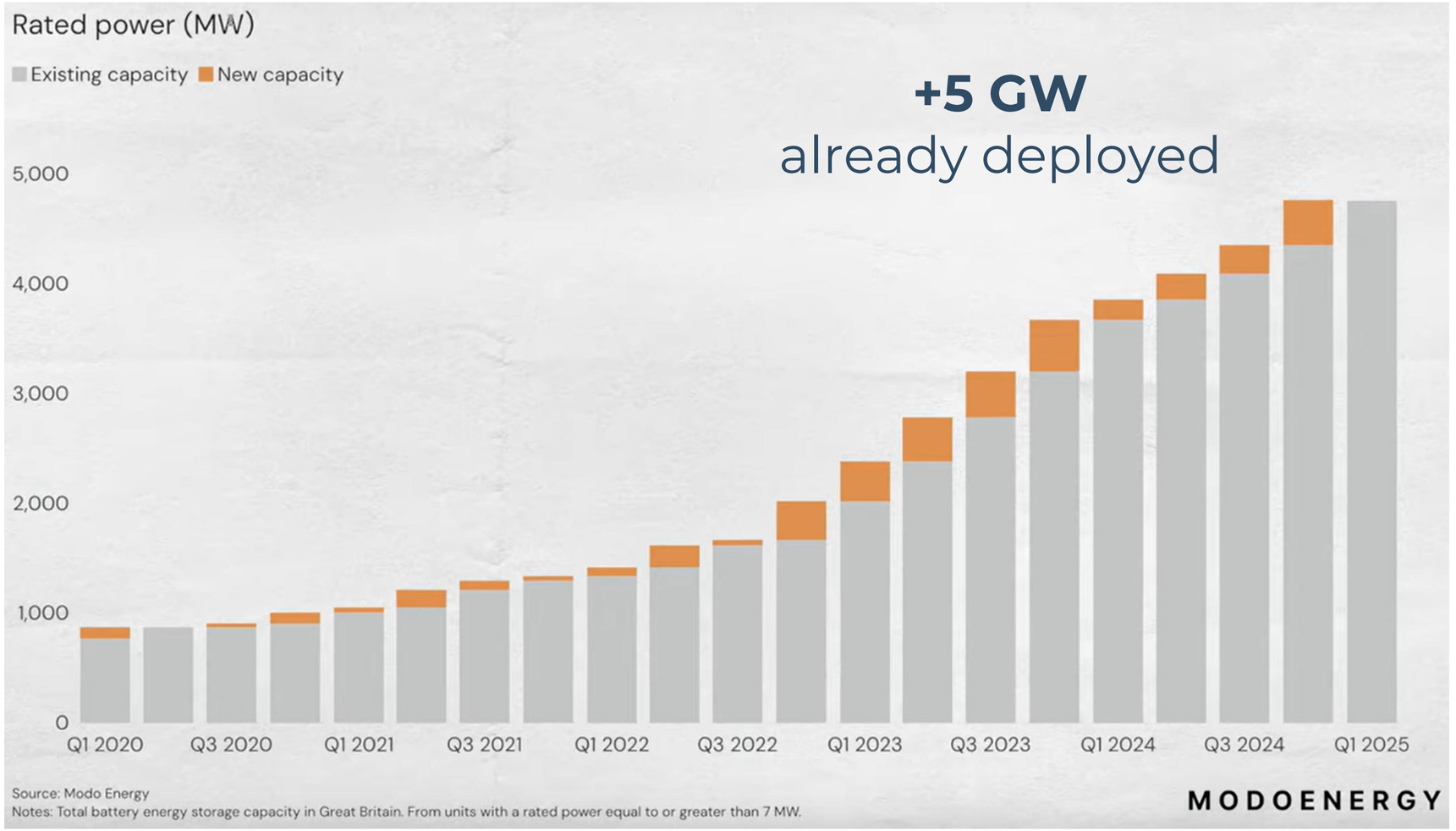
Texas (US)



Australia



Battery storage buildout in GB

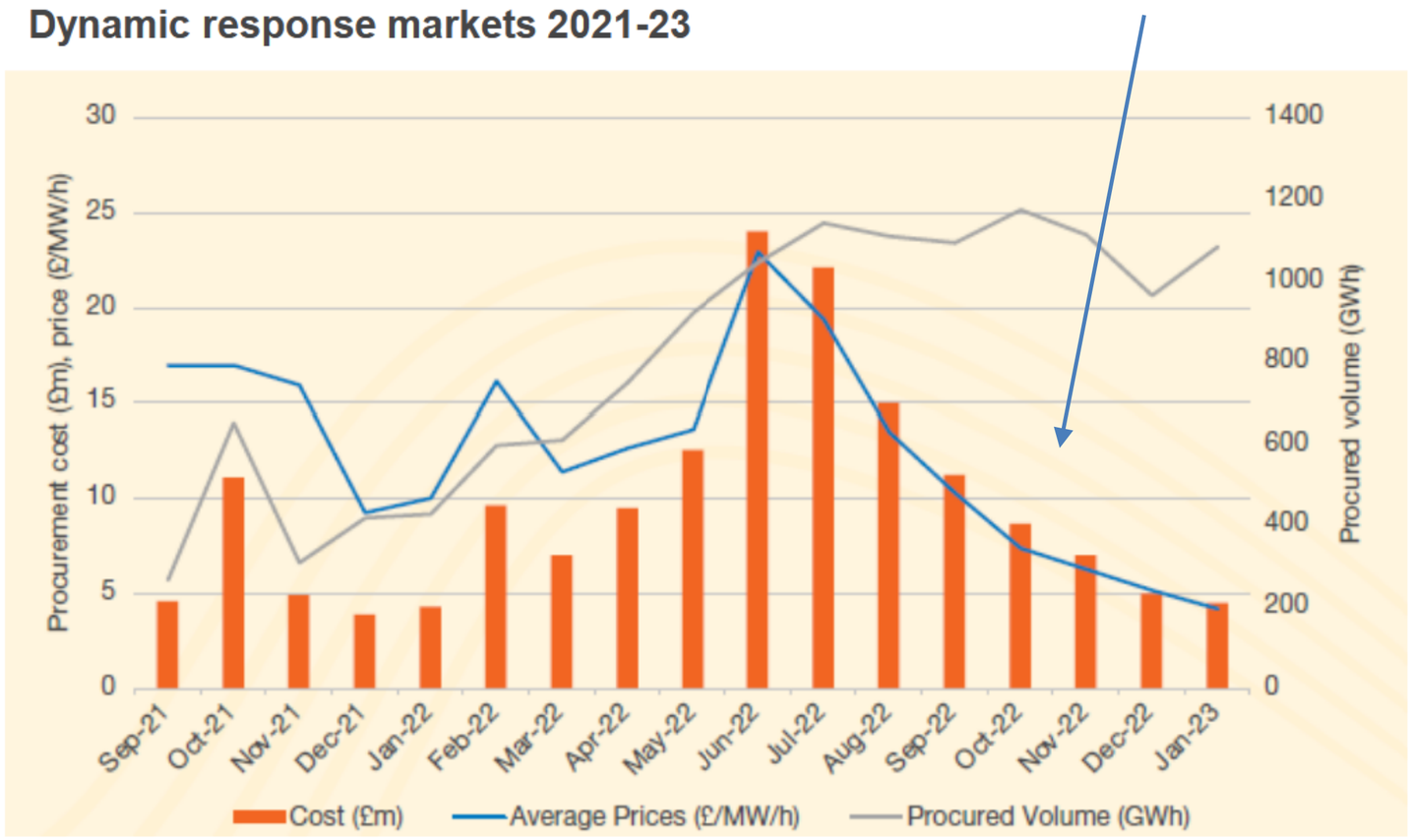


The market for **frequency reserves** was the **original driver** for battery deployment

However, the **frequency** reserves **market** is already **saturated** (service requirement in GB is < 1.2 GW)

Battery storage buildout in GB

Sharp **drop in prices** since 2022



The market for **frequency reserves** was the **original driver** for battery deployment

However, the **frequency** reserves **market** is already **saturated** (service requirement in GB is < 1.2 GW)

Source: National Grid ESO, 2023 Markets Roadmap webinar slides

New market: synthetic inertia

nationalgridESO

~£650m for
Phases 1 and 2



Stability Pathfinder

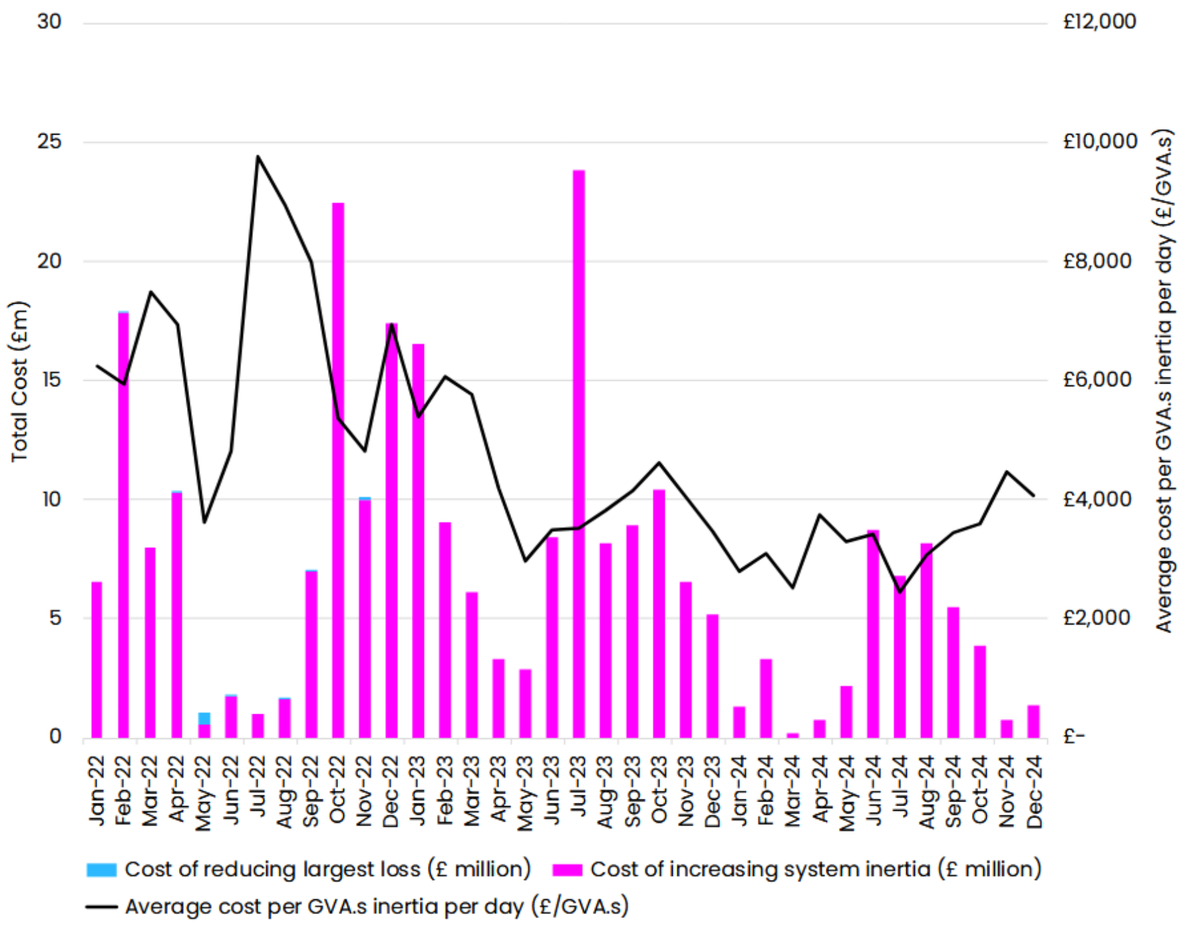
- Tenders took place during **2020 – 2022**
- **Long-term contracts** for **inertia** and Short-Circuit Level providers
  - Inertia and SCL services were decoupled, opening the door to power electronics which cannot provide much SCL
- **Technology agnostic**
  - **In practice**, only synchronous condensers and **batteries + GFM** were eligible
  - Required delivery of a sustained service (weather-driven assets such as wind and solar PV excluded)

New market: synthetic inertia

Stability Pathfinder

- Has achieved a **reduction** in **running costs** for inertia management since 2024

ST Figure 3: Inertia management costs Jan 2022 - Dec 2024



Source: NESO, Markets Roadmap, March 2025

Why these inertia and fast reserve markets?

GB is a **weak** system

theguardian Fri 9 Aug 2019

Transport chaos across England and Wales after major power cuts



What happened in **2019**?

- Loss of a windfarm and a gas plant due to lightning strikes (***N*-2 event**)
- **1.1 million electricity customers** were without power for between 15 and 50 minutes
- Significant **disruption** in the **rail network**

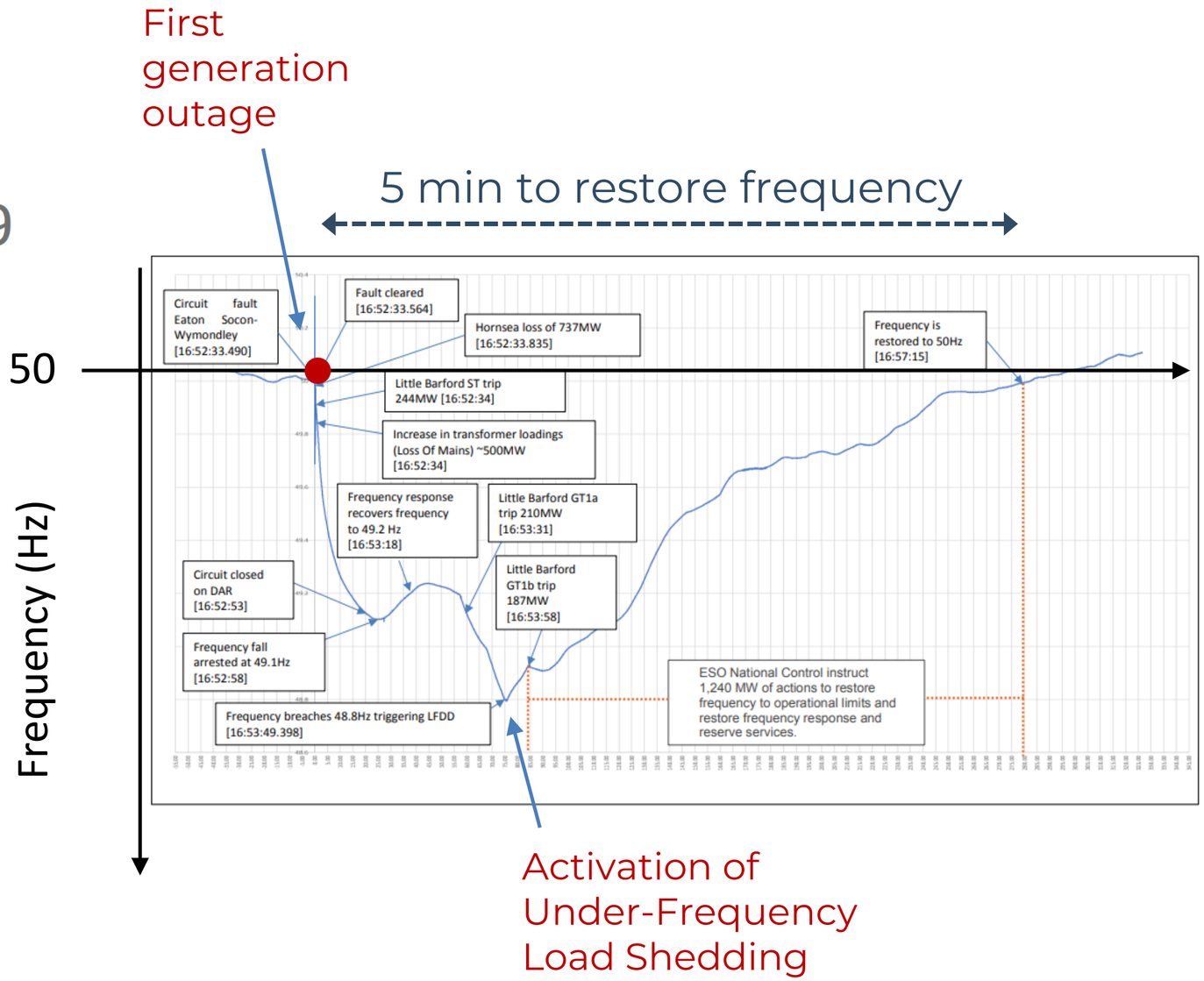
Why these inertia and fast reserve markets?

GB is a **weak system**

theguardian Fri 9 Aug 2019

Transport chaos across England and Wales after major power cuts





Source: National Grid report on the 2019 incident

Why these inertia and fast reserve markets?

GB is a **weak system**

During the **COVID-19 lockdown**, supressed demand led to low electricity prices, resulting in this generation mix:

Renewables  +  **Nuclear**

This system **lacked inertia**,
gas-fired plants had to be **turned on** simply for stability!

Inertia-related costs in GB were ~£300m during May to July 2020,
3 times more than in the same period in 2019

Source: L. Badesa *et al.*, "Ancillary services in Great Britain during the COVID-19 lockdown: A glimpse of the carbon-free future", 2021

Why these inertia and fast reserve markets?

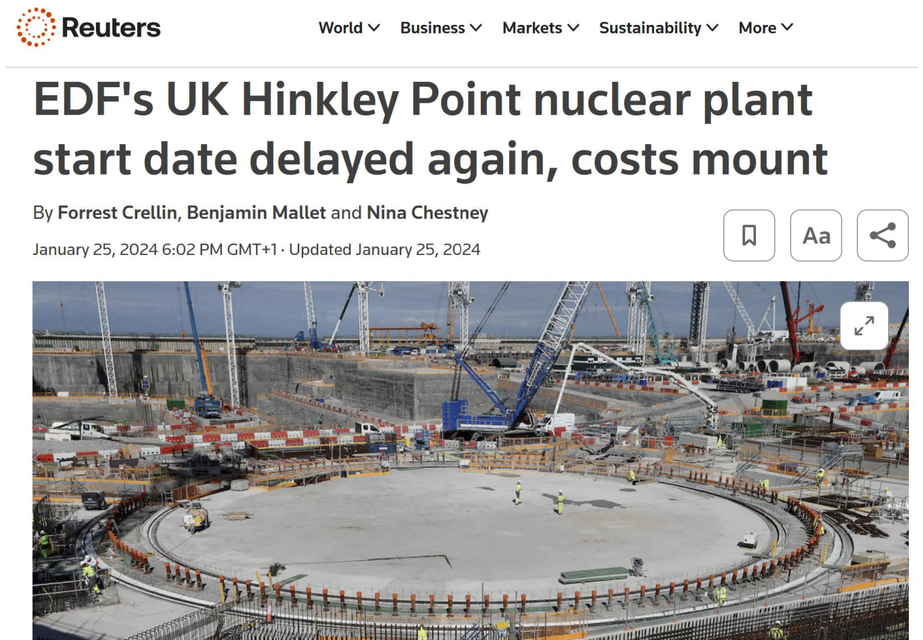
GB has **large *N*-1** contingencies

- Large cross-country **HVDC interconnectors**

2021	2023	2025	2029
North Sea Link @ 1400MW	**Viking Link** @ 1400MW	**Damhead Creek 2** @ 1800MW	**EuroLink** @ 1800MW

Source: National Grid, Operability Strategy report 2022

- Large **nuclear plants**



What's next for BESS supporting stability?

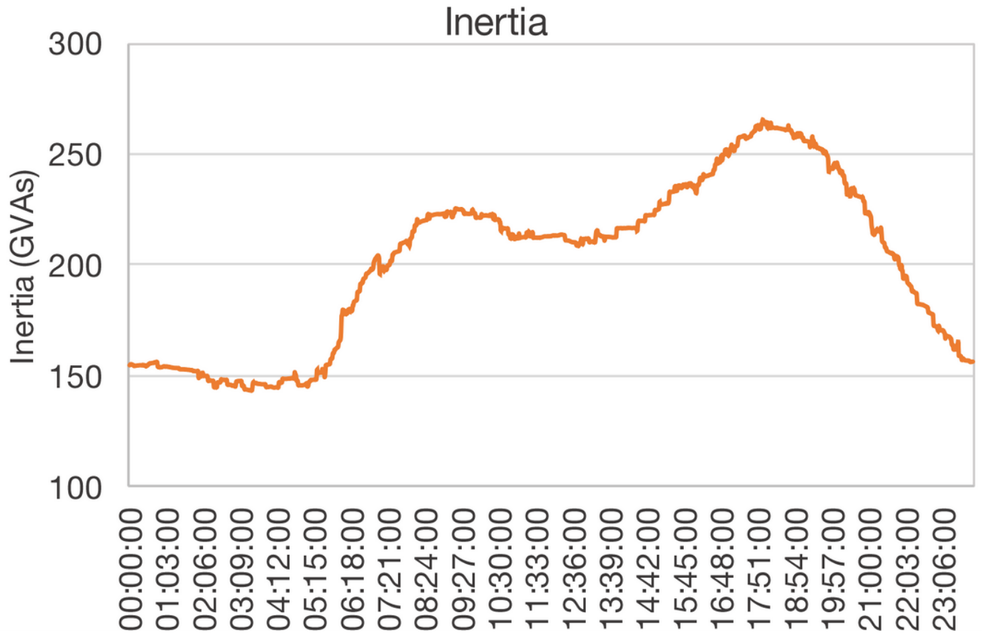
Creation of **short-term markets** for inertia

		Long-term market	Year-ahead[1]	Short-term market
Timeframe	Requirement determination	Annual	Annual	Daily
	Frequency of procurement	Annual	Annual	Daily
	Procurement lead time	T-[5]: Prequalification T-[4] : Procurement ***Industry preference for T-4 based on initial feedback***	T-1: Procurement (if needed to correct forecast error/closures, preferred to structure as a call option)	Day-ahead *TBC exact timing based on ESO internal processes*
	Contract duration	10y/15y/longer ***Industry preference for 10yrs based on initial feedback***	1 year	Daily 23:00 D to 23:00 D+1

Source: AFRY, "Stability market design – Preferred option", Annex report to National Grid ESO, March 2022

What's next for BESS supporting stability?

Creation of **short-term markets** for inertia



Source: National Grid ESO, "Operating a Low Inertia System - A System Operability Framework document", 2020

Typical **daily inertia profile** in GB in 2020:

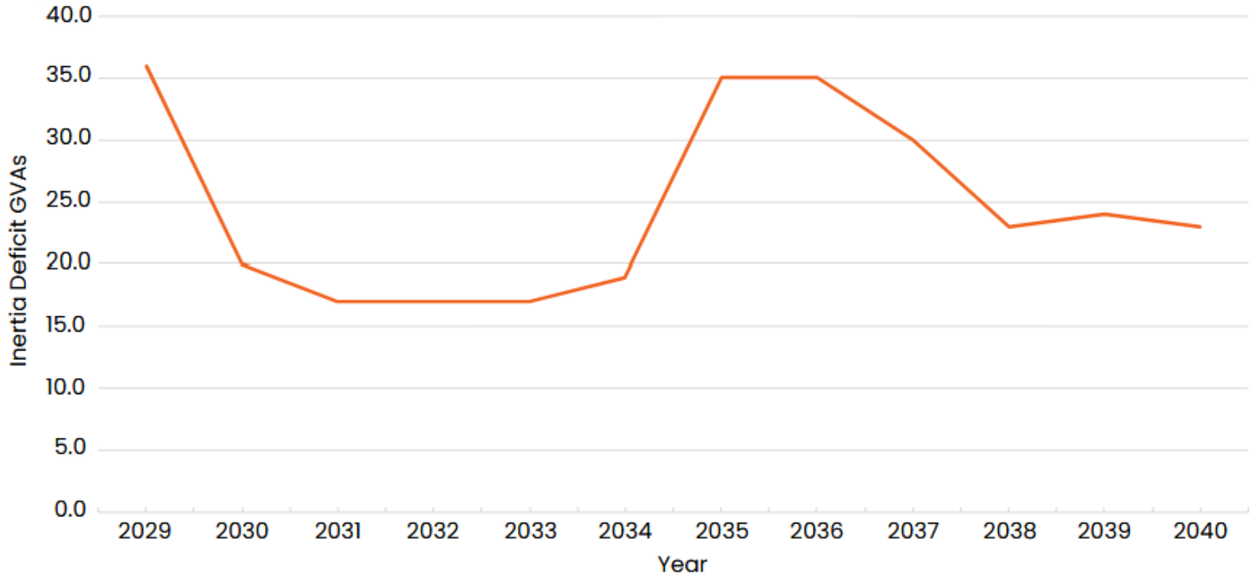
- Inertia is **lowest during night** (when demand is low and wind generation is typically high)

- Enable the system operator to **adjust** procured volumes of **inertia** depending on the **hourly needs**
- Open **participation** to **RES** (otherwise unable to forecast position months in advance) ⟶ **new competitor** for BESS

What's next for BESS supporting stability?

But **inertia** will **remain a concern** in coming decades:

Figure 3: Peak Inertia Deficit GVAs



Source: NESO, Markets Roadmap, March 2025

What's next for BESS supporting stability?

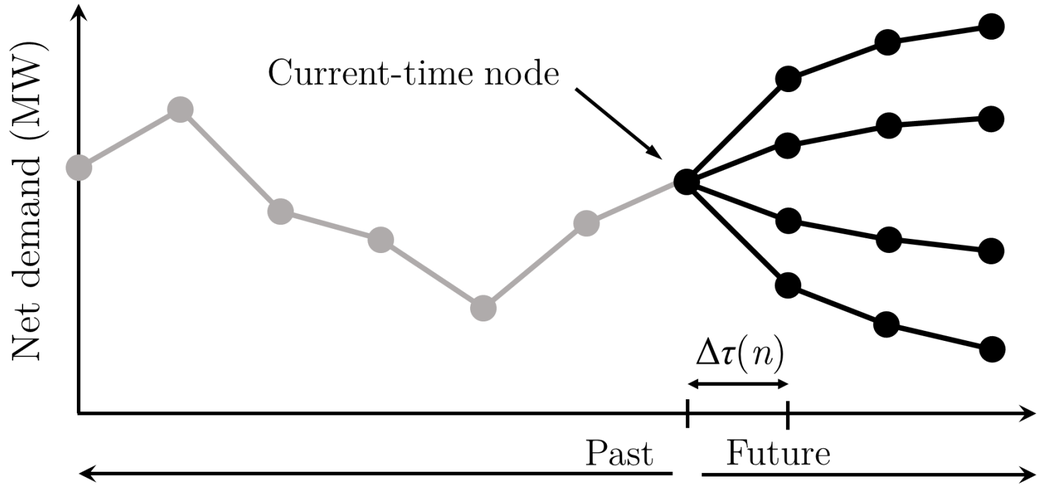
And **new markets** are already envisioned by NESO:

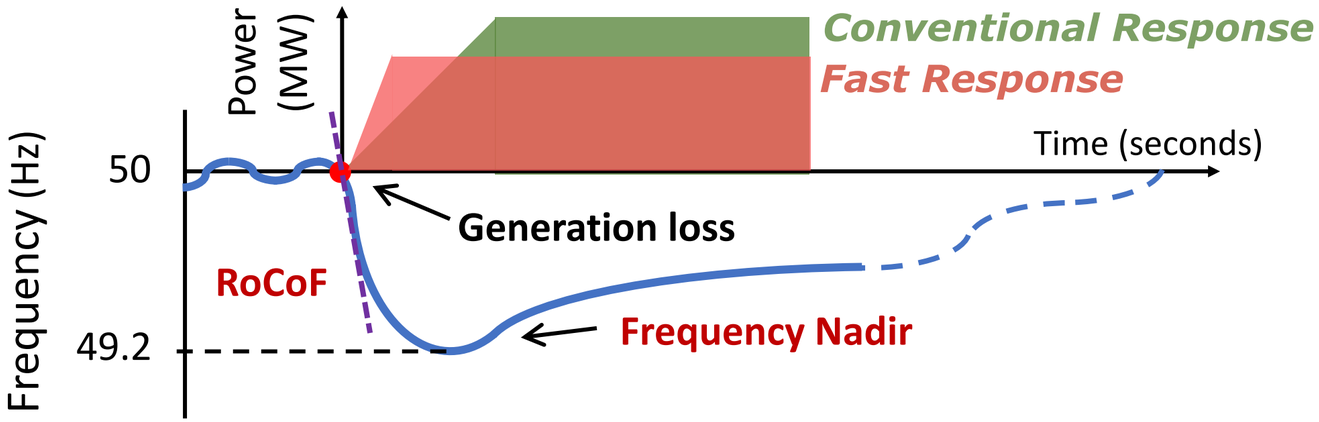
- Voltage support, congestion alleviation…
- **Locational requirements** will likely become much more important
- **Well suited** for **BESS**

More details in NESO's '**Stability markets**' plans

Research in techno-economics of systems with BESS

- Considering uncertainty from RES generation (Stochastic Unit Commitment)
- Guaranteeing frequency stability by scheduling sufficient **ancillary services**





Minimize expected operating cost: fuel costs, start-ups, etc.

$$\min \quad \sum_{n \in \mathcal{N}} \pi(n) \sum_{g \in \mathcal{G}} C_g(n)$$

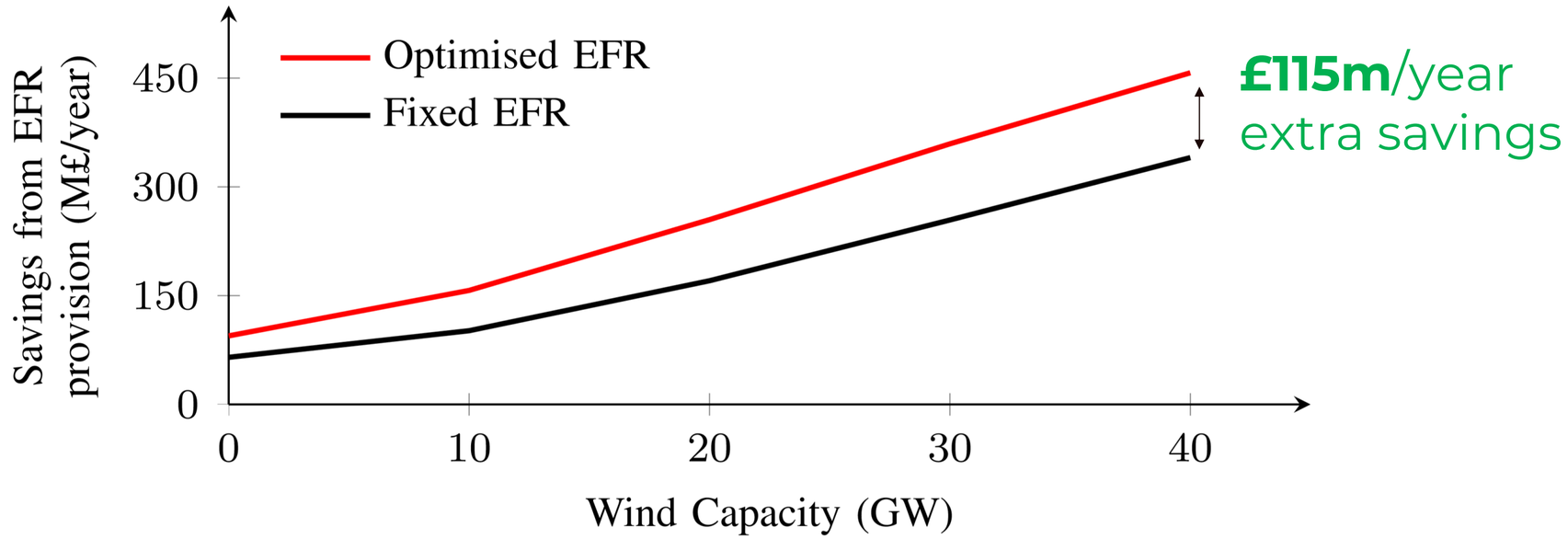
$$\text{s.t.} \quad \begin{array}{l} \text{RoCoF constraint} \\ \text{Nadir constraint} \\ \text{SteadyState constraint} \end{array}$$

Guarantee sufficient ancillary services

(and the rest of the technical constraints)

Results: hourly optimization of BESS operation

- Fast frequency reserve **is not always needed**:
  it only becomes significantly more valuable when inertia is low
- **Co-optimizing EFR procurement** can achieve **savings** of up to **£115m/year**

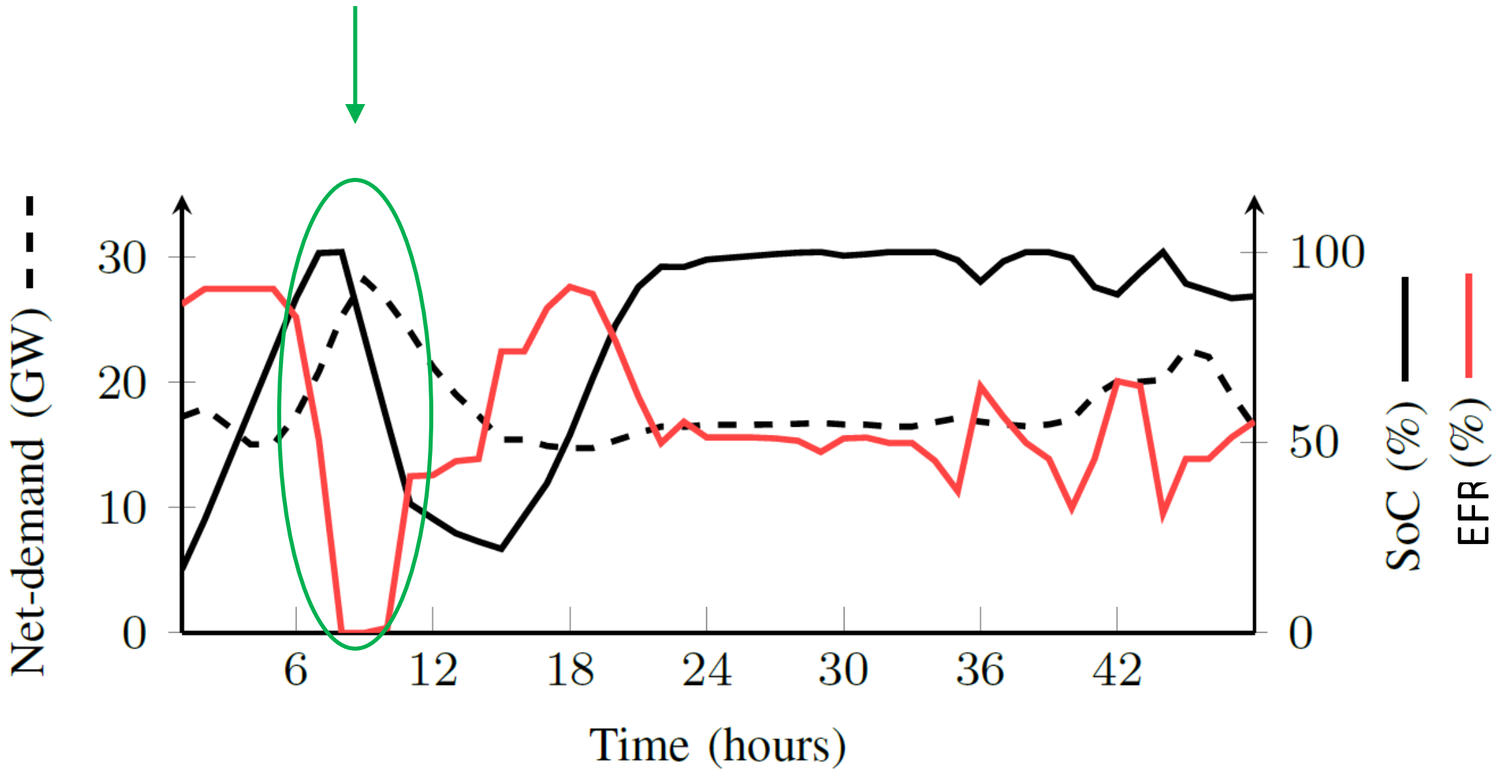


Results: hourly optimization of BESS operation

Where do these extra savings from co-optimizing EFR come from?

EFR is **not needed** when **net-demand is high**:

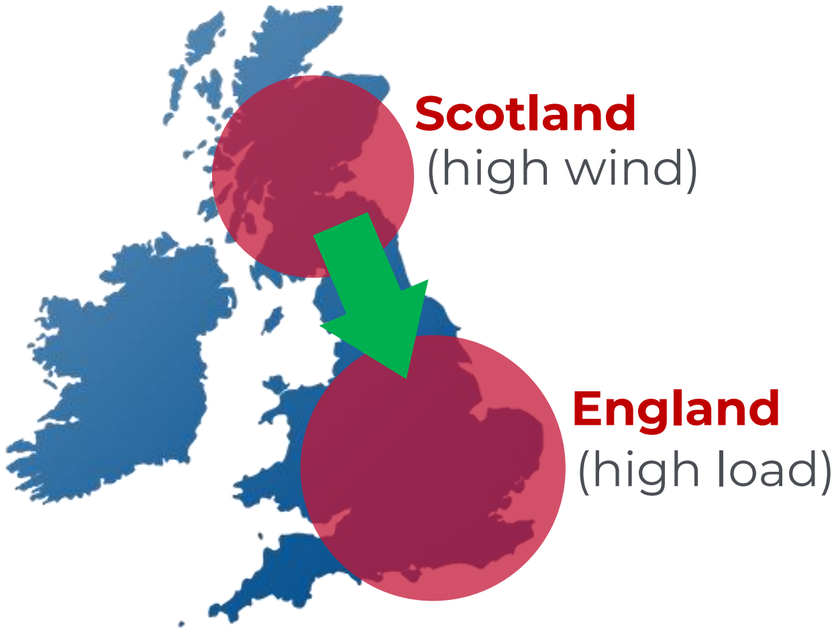
better to use BESS for **energy arbitrage**

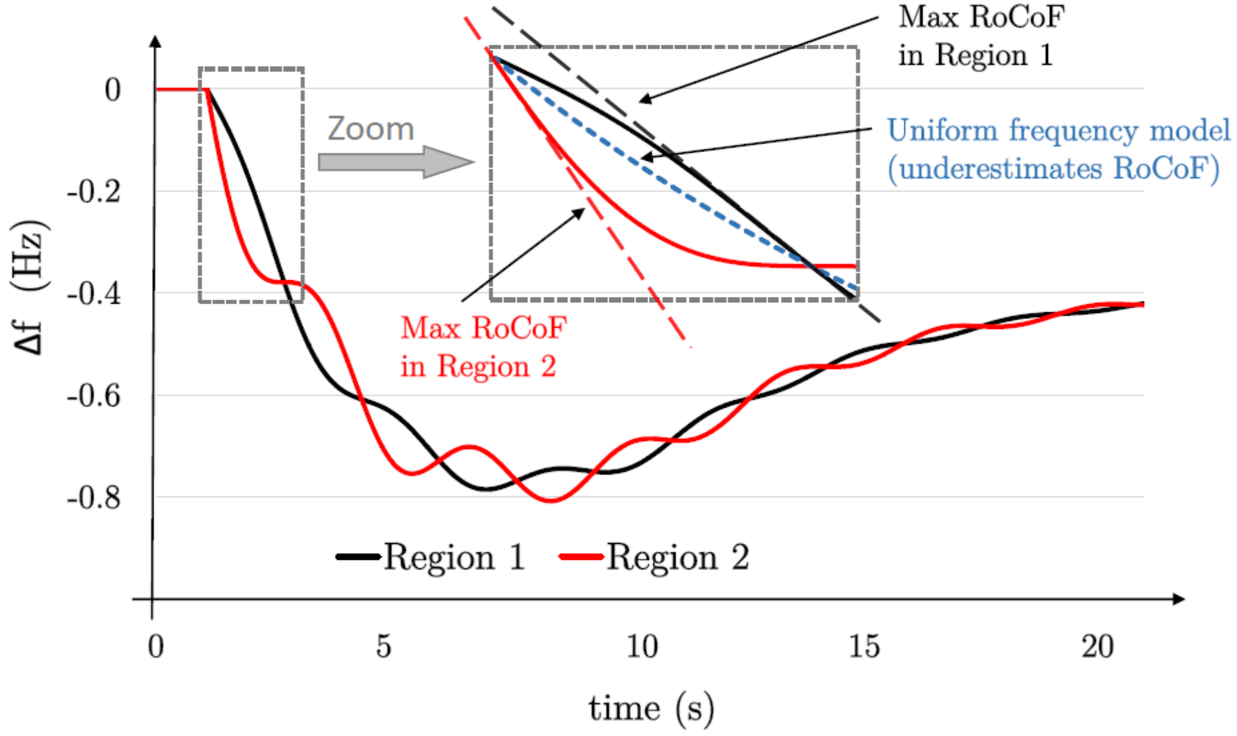


Increasing concern: regional frequency variations

- Inter-area oscillations appear **when inertia is not evenly distributed in the grid** (*e.g.*, high wind in Scotland but most of demand located in England)

- Ignoring **inter-area oscillations** could be **dangerous**: higher RoCoF and lower frequency nadir than the COI could lead to unexpected problems

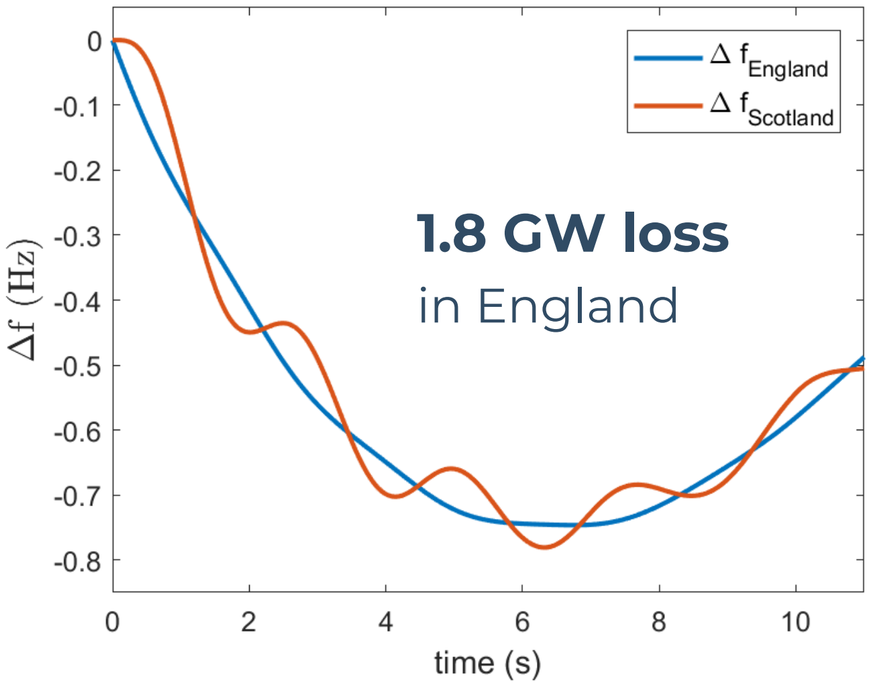


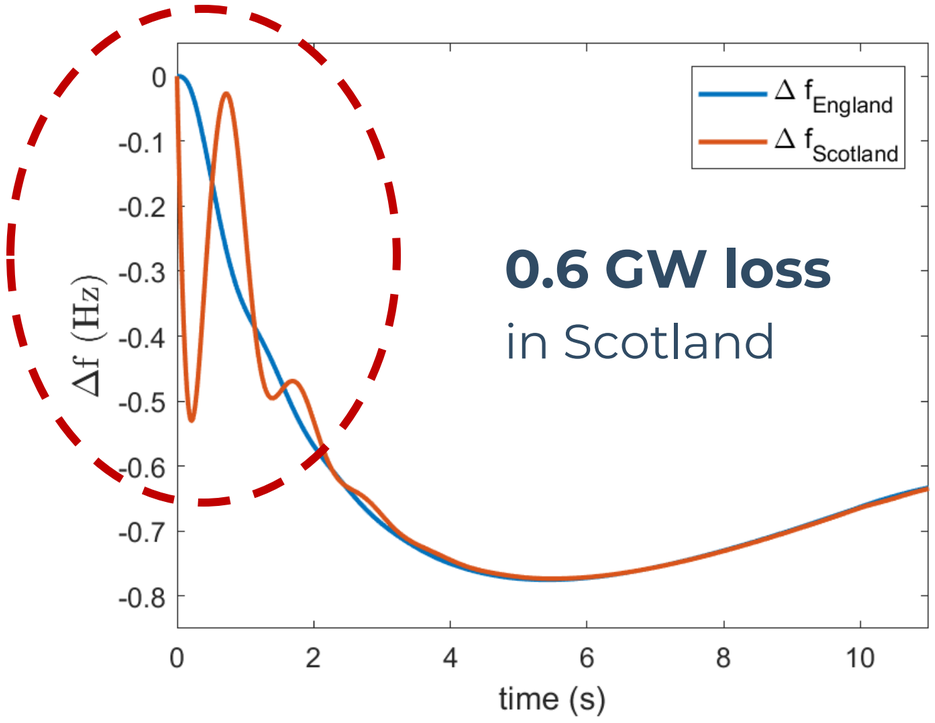


When is the uniform frequency model not accurate?

When there is a **non-uniform** inertia distribution and a **fault** occurs in the **low-inertia region**

Example: **90%** inertia **in England** - **10%** in **Scotland**

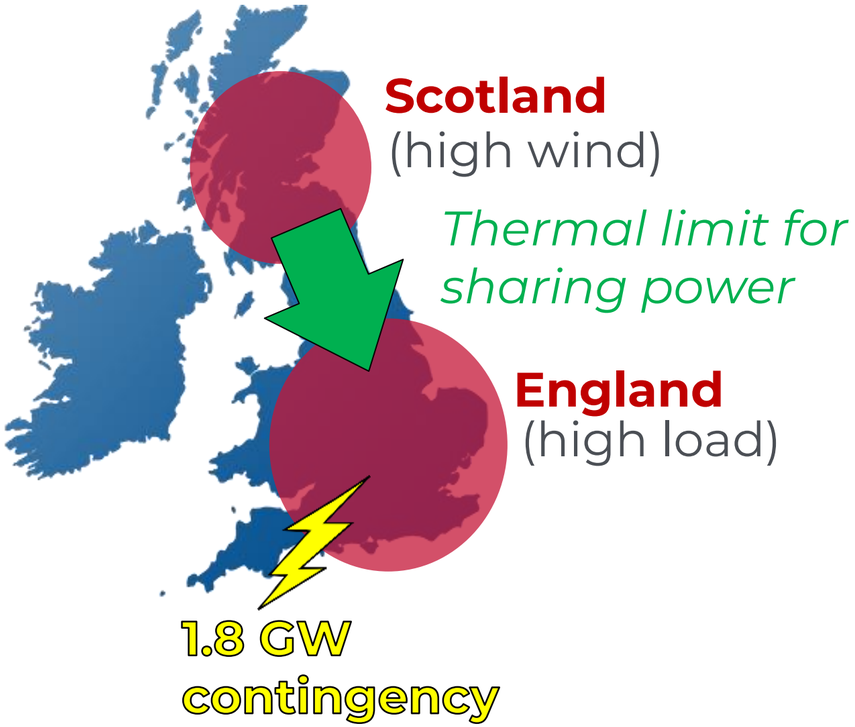


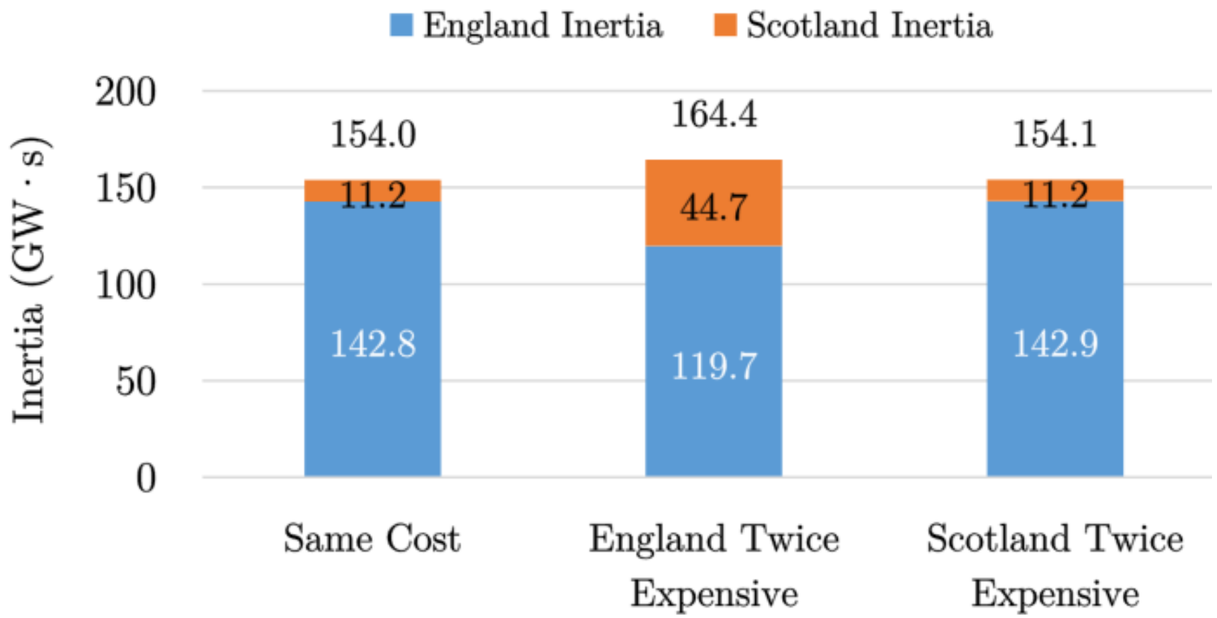


Results: Have some inertia in Scotland!

But exactly how much inertia in England and how much in Scotland?

It **depends on** the **cost**!

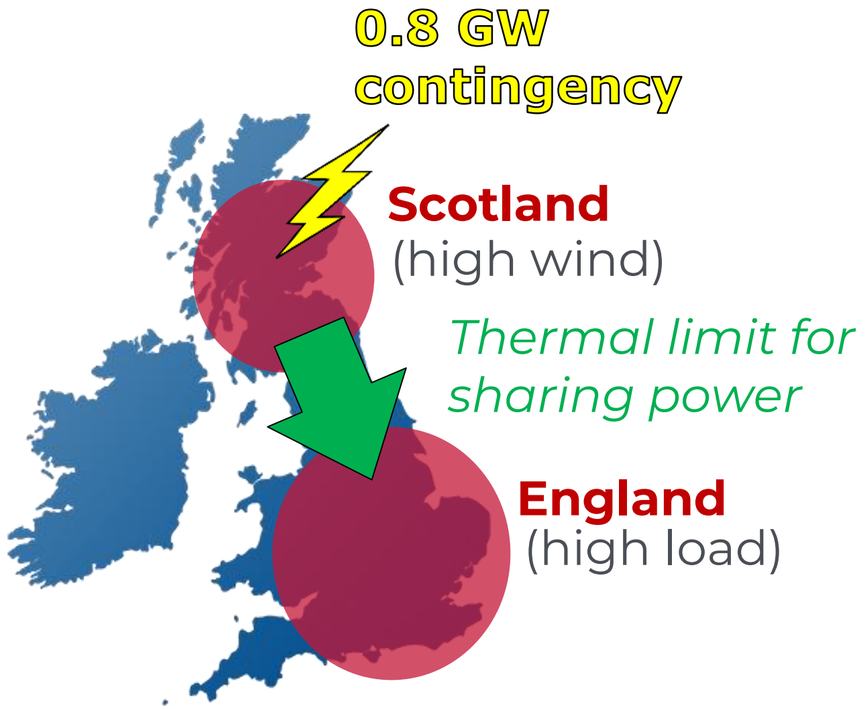


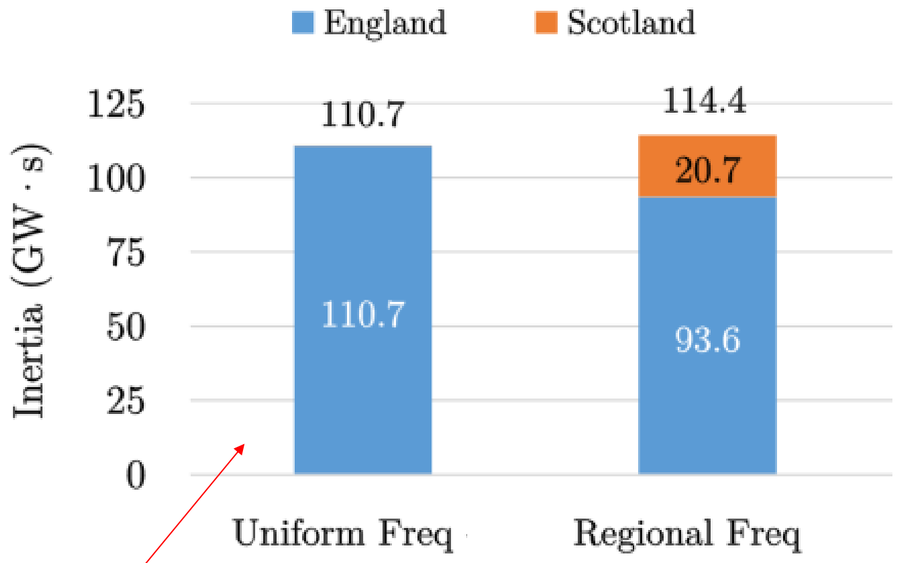


Results: Move to <u>regional</u> *N*-1!

It is key to consider the ***N*-1 contingency** in **<u>each region</u>**!

Assuming that the largest overall contingency in GB is enough to keep the system stable, could lead to **RoCoF violation** in **Scotland**





This case would violate the RoCoF limit in Scotland

Ongoing research at ETSIDI-UPM

Project **BEATLES** (2025 - 2027)

StaBility-ConstrainEd OptimAl DispaTch of Low-Carbon PowEr GridS







Project **DescarbonElec** (2025 - 2026)

Mercados de estabilidad para redes eléctricas descarbonizadas



¡**Gracias** por su atención!